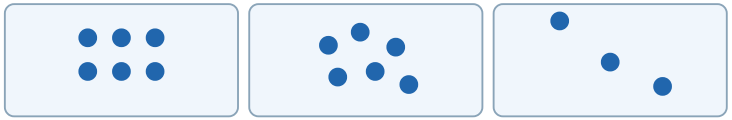
States & density

Model particles

1 Three states



Solid | Liquid | Gas

Solid: close, fixed arrangement; vibration. Liquid: close, irregular, able to flow.

Gas: widely spaced, random motion; easily compressed. Individual particles do not expand when heated.

2 Physical changes

Melting/freezing; boiling or evaporation/condensation; sublimation solid → gas.

No new substance; reversing restores properties. Open-container mass loss may be escaping vapour.

Measure volume

3 Regular & irregular solids

Weigh solid. Cuboid volume = lwh; cylinder = $\pi r^2 h$. Use ruler, callipers or micrometer.

Irregular: fully submerge; final water volume − initial = solid volume. Displacement can is an alternative.

4 Liquids & good readings

Weigh empty dry cylinder, then cylinder + liquid. Subtract to find liquid mass.

Read meniscus at eye level. Avoid bubbles/splashes; unsuitable for solids that dissolve or absorb water.

Calculate & evaluate

5 Density equation

$\rho = m/V$; $m = \rho V$; $V = m/\rho$.

240 g / 80 cm^3 = 3.0 g/cm^3 = **3,000 kg/m^3.**

1 cm^3 = 10^{-6} m^3. Keep units consistent.

6 Explain changes

Fixed mass spreads over larger volume → lower density. Gas spacing explains low density.

Do not claim every solid is denser than its liquid: ice is an exception. Repeats assess random variation; check zero errors.